THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $(a+b)^2$ equals: 1

(a) $2a+2b$ **(b)** $a^2+2ab+b^2$ **(c)** a^2+ab+b^2 **(d)** a^2+b^2

2. The ratio of 50 cm to 2 m is: 1

(a) 4 : 1 **(b)** 2 : 4 **(c)** 1 : 4 **(d)** 1 : 5

3. The price ₹ 500 is increased by 10%. The new price (in ₹) is: 1

(a) 550 **(b)** 495 **(c)** 1100 **(d)** 605

4. The perimeter of a right triangle with legs 3 cm and 4 cm is (in cm): 1

(a) 10 **(b)** 7 **(c)** 12 **(d)** 13

5. On a map, 1 cm represents 5 km. A distance of 7 cm on the map is actually (in km): 1

(a) 12 **(b)** 40 **(c)** 70 **(d)** 35

6. The value of 21^2 is: 1

(a) 441 **(b)** 420 **(c)** 462 **(d)** 451

7. ₹ 400 is shared in the ratio 3 : 5. The larger share is: 1

(a) ₹ 200 **(b)** ₹ 50 **(c)** ₹ 250 **(d)** ₹ 150

8. The simple interest on ₹ 1000 at 5% per annum for 2 years (in ₹) is: 1

(a) 200 **(b)** 90 **(c)** 110 **(d)** 100

9. A rectangle is 3 cm long and 4 cm broad. Its diagonal (in cm) is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 7

10. In a pie chart a sector represents 10% of the data. Its central angle (in degrees) is: 1

(a) 36 **(b)** 30 **(c)** 40 **(d)** 20

11. The value of $3x+5$ when $x=4$ is: 1

(a) 7 **(b)** 20 **(c)** 17 **(d)** 12

12. The cost of 6 pencils is ₹ 48. The cost of 1 pencil is: 1

(a) ₹ 16 **(b)** ₹ 8 **(c)** ₹ 9 **(d)** ₹ 42

13. The marked price of a shirt is ₹ 1000 and the discount is 10%. The selling price (in ₹) is: 1

(a) 1800 **(b)** 900 **(c)** 810 **(d)** 990

14. The longest side of a right triangle is called the: 1

(a) hypotenuse **(b)** altitude **(c)** base **(d)** median

15. 4 workers can finish a job in 12 days. The number of days 8 workers (working at the same rate) need is: 1

(a) 8 **(b)** 20 **(c)** 48 **(d)** 6

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Expand $7(x + 3)$. 2

17. Find x if 5 : 9 = 15 : x. 2

18. Find 12% of ₹ 500. 2

19. In a right triangle the hypotenuse is 17 cm and one leg is 8 cm. Find the other leg. 2

20. 10 men can build a wall in 24 days. In how many days can 8 men build it? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The marked price of a bag is ₹ 1200. A discount of 10% is given. Find the discount and the selling price. 3

22. Find the perimeter of a right triangle whose legs are 5 cm and 12 cm. 3

23. Divide ₹ 1200 among A, B and C in the ratio 2:3:5. 3

24. Use the identity $(a + b)^2$ to find 24^2. 3

OR

5 kg of rice cost ₹ 225. Find the cost of 8 kg of rice. 3

25. Find the simple interest and the amount on ₹ 4000 for 2 years at 5% per annum. 3

OR

Rohan walks 6 km east and then 8 km north. How far is he from his starting point? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The data of 30 students' choices is: Cricket 12, Football 9, Tennis 6, Hockey 3. Find the central angle of each sector for a pie chart and verify that the angles add to 360°. 5

OR

Find 104×25 using the distributive property and verify by direct multiplication. 5

27. 6 pens cost ₹ 90. Find the cost of 1 pen, the cost of 10 pens and the number of pens that can be bought for ₹ 210. 5

OR

A trader buys a table for ₹ 2000. Find the selling price if he sells it at 5

(i) a gain of 10%

(ii) a loss of 5%. Also find the difference of the two selling prices.

END OF QUESTION PAPER — Total: 50 Marks

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